

CIFAR-10

Image *Classification*

A multiclass image classification project using CNNs and ResNet50 transfer learning.

TASK

Multiclass
classification

CLASSES

10 mutually
exclusive labels

INPUT

RGB images
32 × 32 px

TRAINING SET

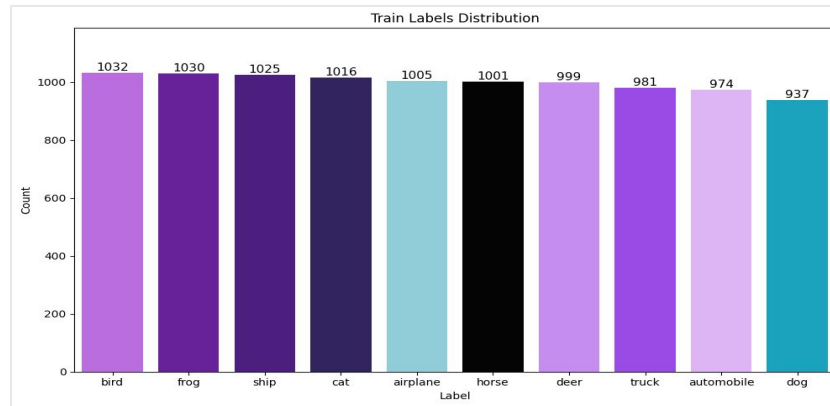
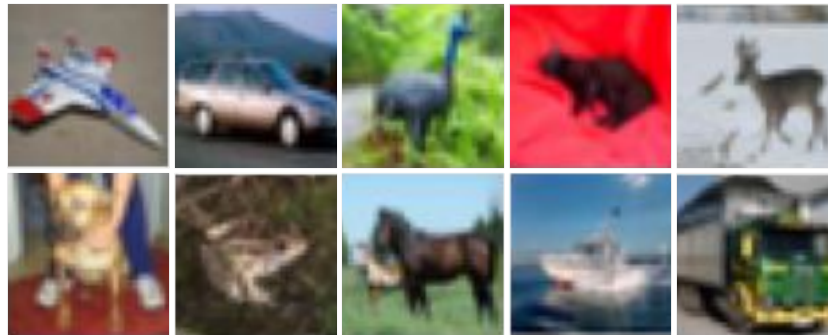
10,000 images
(local hardware)

STEP 01

Exploratory Data Analysis

- 01 Balanced class distribution across all ten labels.
- 02 Strong intra-class variability in pose, background and lighting.
- 03 Low-resolution, blurry, and visually ambiguous samples.
- 04 Different color distributions across classes.

A short look at the dataset before any modelling decisions were made.

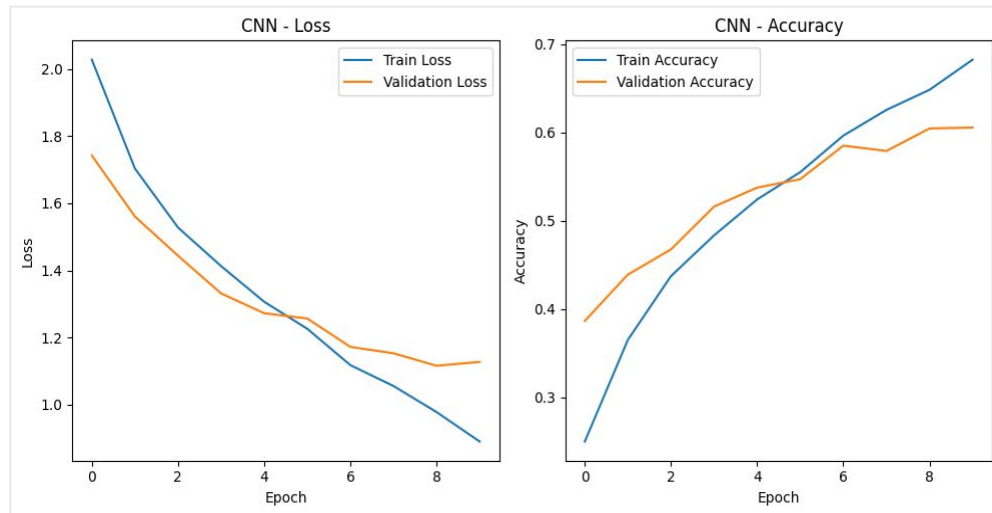


STEP 02

A small convolutional network trained directly on the original 32×32 images.

Baseline CNN

Layer	Output Shape	Parameters
Conv2D + MaxPooling	32×32×32	896
Conv2D + MaxPooling	16×16×64	18k
Conv2D + MaxPooling	8×8×128	74k
Dense + Dropout	128	262k
Output Layer	10 classes	1.3k



Total trainable parameters: ~357k

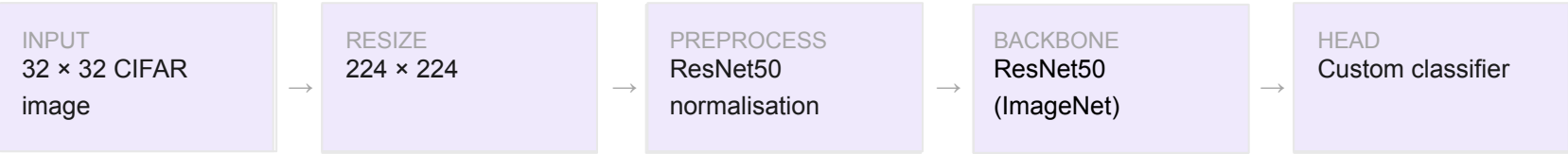
FINAL TRAIN ACCURACY

76%

FINAL TEST ACCURACY

60.5%

ResNet50 Transfer Learning



TRAINING STRATEGY

- 01 Freeze the pretrained backbone, train only the custom classification head.
- 02 Selective fine-tuning of deeper layers with a much smaller learning rate.
- 03 Data augmentation: flip, crop, brightness and contrast variations.
- 04 ReduceLRonPlateau to adapt the learning rate dynamically.

Component	Output Shape	Parameters
Input Layer	224×224×3	0
ResNet50 Backbone	7×7×2048	23.6M
Global Average Pooling	2048	0
Dense Layer	256	524k
Dropout	256	0
Output Layer	10 classes	2.5k

Improving Generalization

DATA AUGMENTATION

Artificially increase visual diversity without changing the labels.

- Random horizontal flip
- Random crop / zoom
- Brightness variation
- Contrast variation



ORIGINAL

FLIP

CROP

BRIGHTNESS

CONTRAST

OVERFITTING CONTROL

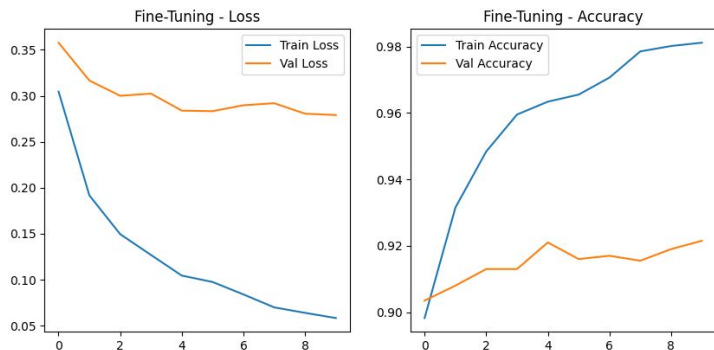
Regularization and adaptive learning rate to stabilize training.

- Dropout regularization
- Selective layer freezing
- Smaller fine-tuning learning rate
- ReduceLROnPlateau callback

GOAL Improve generalization on a relatively small dataset.

Results

MODEL	TEST ACCURACY
Baseline CNN	60.5%
ResNet50 head only	88.3%
Fine-tuned ResNet50	91%



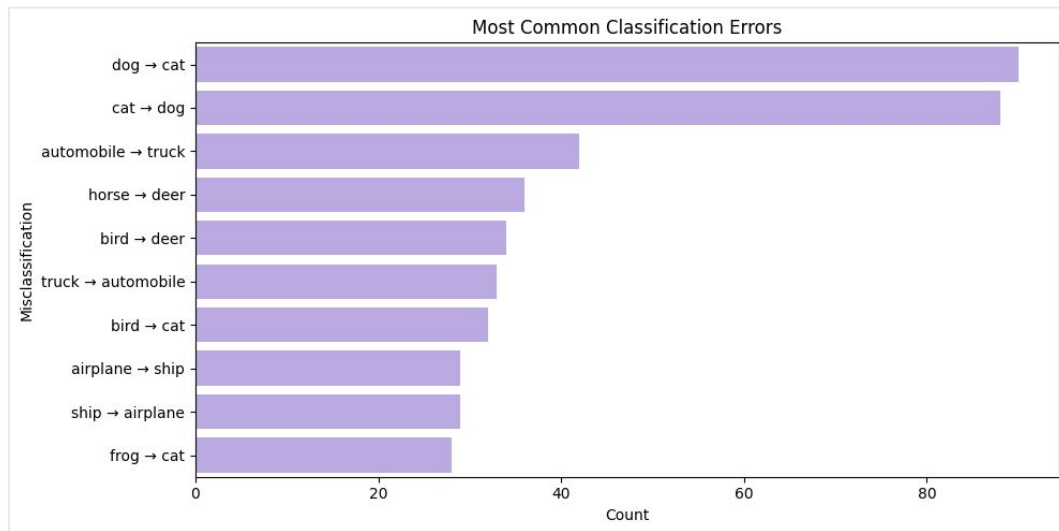
KEY OBSERVATIONS

Transfer learning significantly improved test accuracy compared to the baseline CNN.

Selective fine-tuning further improved performance after the head-only training stage.

Data augmentation and selective freezing helped reduce overfitting and improve generalization.

Error Analysis



- Most errors occurred between visually similar classes
- Some failures were caused by unusual viewpoints or cropped objects
- Low-resolution images still limit fine-grained recognition

True: bird
Pred: deer (0.59)



True: airplane
Pred: dog (0.57)



True: frog
Pred: cat (0.62)



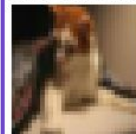
True: bird
Pred: cat (0.72)



True: horse
Pred: cat (0.65)



True: cat
Pred: dog (0.74)



True: automobile
Pred: truck (1.00)



True: deer
Pred: dog (0.57)



True: frog
Pred: cat (1.00)



CLOSING

Conclusion

- Transfer learning strongly outperformed the baseline CNN
- Data augmentation improved generalization
- Low-resolution images remain challenging
- Error analysis revealed meaningful semantic confusions

FINAL TEST ACCURACY

91%

Fine-tuned ResNet50 - 10,000-image subset

Future Work:

- EfficientNet / Vision Transformers
- Stronger augmentation
- Larger training subset

Thank you for listening.